

Phiona B. Umwali

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Education

Cornell University, College of Computing and Information Science

Ithaca, NY

Master of Professional Studies in Information Science

May 2026

Bachelor of Arts in Information Science

May 2025

Concentrations: Data Science · Information Ethics, Law and Policy | GPA: 3.8 / 4.0

Relevant Coursework: Retrieval Augmented Generation (RAG) and AI Agents/chatbots, Computational Text Analysis (ML), Design and Programming for the Web, Ubiquitous Computing

Technical Skills

Programming: Python, SQL, R, JavaScript (Vue.js), HTML/CSS, Pandas, NumPy, scikit-learn, LangChain

Analytics & Visualization: Tableau, Jupyter Notebooks, RStudio, NVivo, Azure Machine Learning, WhereScape

Tools & Technologies: Git, REST APIs, UiPath, Retrieval-Augmented Generation (RAG), Prompt Engineering

Core Competencies: Machine Learning, Predictive Modeling, Statistical Analysis, A/B Testing, Data Visualization, ETL Development, Workflow Automation (RPA), Thematic Analysis, Digital Analytics, Business Analysis.

Relevant Experience / Projects

Quantitative Analyst (Team lead) | MPS Project Practicum: The Riverside Company Aug 2025 - Dec 2025

- Designed and built logistic regression models using Python (Pandas, scikit-learn) to analyze underwriting projections (ION Sales/EBITDA) across 100+ portfolio companies, leading to statistical analysis that identified early underperformance signals in 30+ companies and found that companies missing projections by 50% were 60% more likely to generate negative IRRs.
- Implemented automated ETL pipelines to clean, reconcile, and normalize multi-year company performance and forecast data, generating Tableau-ready datasets for investment performance analysis and reporting.

Data Analytics Research Assistant | BALTO Community Data Labs, University of Maryland May 2025 - Dec 2025

- Designed an NVivo coding framework to analyze 50+ community interviews and workshop transcripts, transforming unstructured qualitative data into standardized, theme-tagged datasets.
- Synthesized findings in R and Tableau to identify transit accessibility and service reliability challenges, producing reports that informed community advocacy and policy recommendations.

AI Research Engineer | Cornell AI Lab, Cornell Center for Law & AI Aug 2025 - Dec 2025

- Developed and evaluated two RAG architectures in Python, a LangChain/FAISS chunked retrieval pipeline and a long-context retrieval system with document compression, benchmarking retrieval accuracy, citation quality, and response relevance across 200+ legal queries.
- Built an end-to-end conversational AI pipeline with dynamic prompt engineering, retrieval context injection, and structured response generation; tracked token usage and cost-per-query metrics to optimize performance.
- Designed IRB-compliant research infrastructure using Python, HTML, and REST APIs, implementing consent workflows, feedback collection, and A/B testing to evaluate alternative RAG architectures.

Course Teaching Assistant | Cornell Bowers College of Computing and Information Science Aug 2024 - Dec 2025

- Led Python, SQL, and analytics lab sessions for 100+ students, providing technical mentorship and code reviews for projects involving machine learning, databases, data pipelines, and AI applications.
- Designed and evaluated assessments on machine learning, algorithmic bias, data ethics, and AI system tradeoffs, grading 100+ student projects and delivering actionable feedback on technical implementation and impact.

Digital Engagement Team Lead | Responsible AI Network Feb 2024 - Dec 2025

- Directed a data-driven engagement strategy across LinkedIn, Instagram, and X, increasing audience engagement by 40% through analytics-driven content optimization.

- Created content on AI governance, algorithmic bias, transparency, and responsible AI, translating technical and policy concepts for diverse audiences.
- Executed monthly dashboards tracking engagement, audience growth, and content performance, identifying trends to inform outreach, advocacy, and event programming.

Automation Business Analyst Team Lead | UiPath Student Developers

Feb 2024 - May 2025

- Delivered RPA training workshops for 20+ members using UiPath Studio, covering workflow design, UI automation, exception handling, and enterprise automation frameworks.
- Translated business requirements into technical specifications and process definition documents (PDDs), identifying automation opportunities and supporting 10+ developers through bot development, user acceptance testing (UAT), and deployment.

Leadership & Community Engagement

- **Outreach Chair, Cornell Office of Student Government** - Drove outreach campaigns to increase student participation in governance programs; coordinated cross-campus messaging across email, social, and event channels.
- **Director of Marketing and Publicity, East African Students Together** - Managed end-to-end marketing and promotional strategy for cultural events, growing event attendance and community visibility.